

JOSHITHA RAJASEKARAN

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LinkedIn | GitHub | Publication

Professional Summary

Computer Science Engineering graduate with a foundation in software development, backend development concepts, cloud technologies, and web application development. Familiar with **Node.js, NestJS, REST API development, Java, Python, SQL, and AWS** through academic work, internships, and hands-on learning. Built projects in backend application development, machine learning, and web-based applications while strengthening problem-solving, debugging, and software engineering fundamentals. Seeking opportunities in **Node.js development, NestJS development, REST API development, backend development, and software engineering roles**.

Technical Skills

Programming Languages: Java, Python, JavaScript, C, SQL

Backend Development: Node.js, NestJS, REST API Development, CRUD Operations, API Routing, Request/Response Handling

Web Technologies: HTML5, CSS3, JavaScript, Flask

Cloud Technologies: AWS EC2, AWS S3, IAM, CloudWatch

Machine Learning / Libraries: TensorFlow, OpenCV, NumPy, Pandas, SciPy, Librosa

Tools & Platforms: Git, GitHub, VS Code, Postman

Core Subjects: Data Structures, Algorithms, DBMS, Operating Systems, Computer Networks, OOPs

Projects

Student Management REST API

Node.js, NestJS, REST API, SQL

- Developed a backend **CRUD application** to manage student records using **Node.js** and **NestJS**.
- Implemented REST API endpoints to **create, retrieve, update, and delete** student data with structured request and response handling.
- Designed backend modules using controllers and services to organize routing, business logic, and data operations.
- Tested API endpoints and validated request flow for student record management and database interaction.

Speech-to-Text Transcription System

Python, TensorFlow, Flask, Librosa, SciPy

- Built a speech-to-text transcription application using **Python, TensorFlow, and Flask** for audio processing, model integration, and web-based interaction.
- Improved transcription accuracy by **18%** using noise augmentation and feature engineering techniques.
- Processed **500+ audio samples** using MFCC and spectrogram-based feature extraction for model training and evaluation.
- Developed a web interface to support real-time transcription workflow and result display.

Smart Accident Detection and Emergency Human Injury Alert System

Python, TensorFlow, OpenCV, NumPy

- Developed a CNN-based accident detection system to identify accident scenarios from image inputs and support emergency response workflows.
- Achieved **92% detection accuracy** on **1000+ labeled images** using image preprocessing and model training techniques.
- Implemented alert workflow logic for accident severity response use cases.
- Worked on image processing, model evaluation, and dataset preparation using **OpenCV** and **NumPy**.

Experience

Cloud Computing Intern

Orionshift, Chennai

Jul 2025 – Aug 2025

Internship

- Managed and monitored **AWS EC2 instances** and **S3 buckets** to support deployment and cloud environment operations.
- Worked with **IAM** and **CloudWatch** for access control, monitoring, and basic cloud infrastructure management.
- Assisted in troubleshooting cloud environment issues and maintaining service availability.
- Prepared infrastructure documentation and setup notes to streamline deployment and support activities.

Full Stack Web Development Intern

Skill Intern

2024

Internship

- Developed responsive web pages using **HTML, CSS, and JavaScript** with focus on clean structure, usability, and maintainable frontend implementation.
- Worked on frontend-backend integration concepts and gained exposure to web application flow, deployment basics, and development lifecycle practices.
- Used **Git** and **GitHub** for version control, collaboration, and source code management.

Publication

Smart Accident Detection and Emergency Response System with Accident Severity Classification

International Journal of Advanced Research in Science, Communication and Technology (IJARSCT), Vol. 6, Issue 1, May 2026

Authors: S. Madhan, Joshitha R, Pravina M, Nithyasri K

[View Publication](#)

Education

B.E. in Computer Science and Engineering

University College of Engineering, Thirukkuvalai

2022 – 2026

CGPA: 8.3 / 10.0

Certifications

- Cloud Computing Internship – Orionshift
- Full Stack Web Development – Skill Intern
- Web Development Internship Certificate
- Academic Rank Holder – Computer Science Engineering